

Correlation Analysis

Hypothesis Testing: Evaluating Correlations

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POLS 222: Slide Deck #18

We can test the significance of relationships between categorical variables using crosstabs with chi-square tests.

Remember that we have some additional options we can use to get more information about the *strength* of those relationships...

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Measures of Association!

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Most measures of association are **proportional reduction in error (PRE)** tests, which means that they tell us how much of the variation in the DV is explained by the IV.

Measures of association:

- **Lambda (λ):** nominal by nominal/ordinal. **PRE. Asymmetrical.**
- **Cramér's V (V):** nominal by nominal. **Not PRE. Symmetrical.**
- **Somers' d (d):** ordinal by ordinal. **PRE. Asymmetrical.**
- **Tau-b (τ_b):** ordinal by ordinal (square). **PRE. Symmetrical.**
- **Gamma (γ):** ordinal by ordinal (rectangle). **PRE. Symmetrical.**

Consider the shape of the data: when the two variables have the same number of categories (e.g., 3×3), it is square. If they have different numbers of categories (e.g., 4×3), it is rectangular.

We already know that “Correlation does not always mean causation...”

But it's important to establish correlation before making causal arguments:

- Recall that **correlation** is the extent to which there is a mutual relationship between variables.
- Do the variables move together in somewhat of a consistent pattern (positive, negative, linear, monotonic, etc.)?

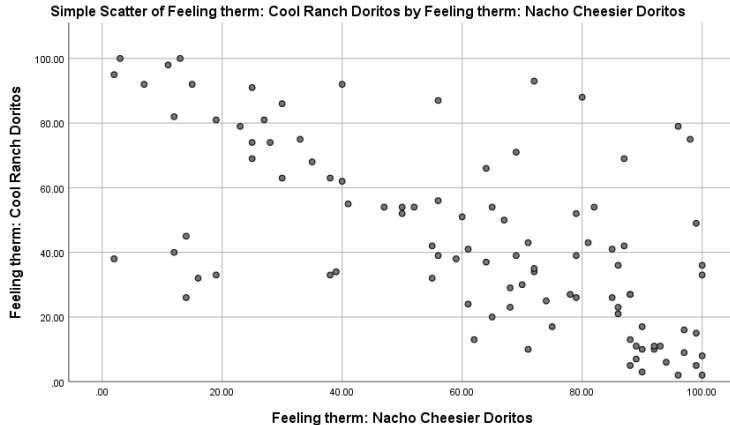
We can sometimes see correlations intuitively, but usually we rely on math to see the strength and direction of the relationship.

Scatterplots are really useful for seeing trends in the data.

- Those trends are correlations.
- Scatterplots align each observation of one variable to the observation of another variable.

For example, if someone didn't really like Nacho Cheesier Doritos (12) but they *did* really like Cool Ranch Doritos (82), we could plot that point at 12 along the X-axis, and at 82 along the Y-axis.

Maybe we can look at all of the feelings toward Doritos and immediately notice a trend in the relationship visually, like in the next slide...



Sure, there are some outliers, but there is generally a negative trend, right?

As people's feelings toward Nacho Cheesier Doritos (the X-axis) increase, their feelings toward Cool Ranch Doritos (Y-axis) tend to decrease.

When we talk about mathematical correlations, we're usually talking about **Pearson's r** .

Pearson's r correlations test the relationship between pairs of observations. We can test a bunch of variables at once, but we will have multiple correlations, one for each pair of variables.

- **H_0** : there is no relationship between the variables
- **H_1** : there is a relationship between the variables

Correlation **is best used between two continuous/interval variables**, but we can technically correlate any variable (and later, we'll learn why we might).

Interpreting Correlations

Correlation tests are symmetrical:

- Correlation doesn't care which variable is the DV and which is the IV; the calculation is the same.
- The “correlation coefficient” indicates how strong the relationship is.
- Every variable will always correlate with itself with a coefficient of positive 1! It's a perfect correlation!

Interpreting Pearson's r pretty straightforward:

- 0.0 to 0.3: weak
- 0.3 to 0.7: moderate
- 0.7 to 1.0: strong
- But it goes from -1.0 to +1.0, so we also look at direction!